

A Signed- M -Matrix Subcase of the Strong Gaussian Correlation Conjecture

Literature-implied partial-answer packet for arXiv:2407.15684
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11 August 2026

Status: literature-implied answer (partial subcase). Karlin–Rinott’s known MTP_2 criterion for absolute Gaussian vectors implies an inequality stronger than the source conjecture whenever the precision matrix is sign-similar to an M -matrix. This implication was identified here; the full conjecture remains open.

1 Source question

Assouline, Chor, and Sadoovsky [1] formulate Tehranchi’s strong Gaussian correlation conjecture in two equivalent forms. For origin-symmetric convex sets $K, T \subseteq \mathbb{R}^d$, the geometric form asks for

$$\gamma_d(K + T)\gamma_d(K \cap T) \geq \gamma_d(K)\gamma_d(T). \quad (1)$$

The Gaussian-box form says that, for any centered jointly Gaussian vector $X = (X_1, \dots, X_n)$ and threshold vectors $s, t \in [0, \infty]^n$,

$$\mathbb{P}(|X| \leq s + t)\mathbb{P}(|X| \leq s \wedge t) \geq \mathbb{P}(|X| \leq s)\mathbb{P}(|X| \leq t). \quad (2)$$

Here all vector operations and inequalities are coordinatewise. The source proves special cases, including unconditional convex sets, but leaves the general assertion conjectural.

The Gaussian Correlation inequality, which is a generalization of the Šidák-Khatri inequality, was proved by Royen in [14] and states the following:

$$\Pr(|X_i| \leq 1 \ \forall i \in [n]) \geq \Pr(|X_i| \leq 1 \ \forall i \in [k]) \Pr(|X_i| \leq 1 \ \forall i \in [n] \setminus [k]). \quad (3)$$

In attempt to generalize Theorem 1 we present the following conjecture, which is equivalent to a conjecture of Teheranchi [19] (stated below).

Conjecture 2. *Let $X_1, \dots, X_n$ be zero-mean jointly Gaussian real random variables, and let $s_1, \dots, s_n, t_1, \dots, t_n \in [0, \infty) \cup \{\infty\}$. Then,*

$$\begin{aligned} & \Pr(|X_i| \leq s_i + t_i \ \forall i \in [n]) \Pr(|X_i| \leq \min\{s_i, t_i\} \ \forall i \in [n]) \\ & \geq \Pr(|X_i| \leq s_i \ \forall i \in [n]) \Pr(|X_i| \leq t_i \ \forall i \in [n]). \end{aligned} \quad (4)$$

Our proof relies on the geometric interpretation of the inequality with respect to the usual Gaussian probability measure in $\mathbb{R}^d$, denoted henceforth as γ_d . The classical Šidák-Khatri inequality can be interpreted as the inequality $\gamma_d(K \cap T) \geq \gamma_d(K)\gamma_d(T)$ when K is an origin-symmetric convex set and T is a symmetric slab. In this geometric setting, this is due to Giannopoulos [6] and Szarek and Werner [18].

Theorem 1 is restated and proved in the geometric setting as follows.

Theorem 3. *Let $K \subseteq \mathbb{R}^d$ be an origin-symmetric convex body and $T = \{x : |\langle x, u \rangle| \leq 1\} \subset \mathbb{R}^d$ a symmetric slab with $u \in \mathbb{R}^d$. Then,*

$$\gamma_d(\text{conv}\{K \cup T\})\gamma_d(K \cap T) \geq \gamma_d(K)\gamma_d(T). \quad (5)$$

Equation (5) does not hold for all origin symmetric and convex $K, T \subseteq \mathbb{R}^d$, as will be seen in Example 14. However, the following weaker inequality, which is a strengthening of Royen's Gaussian correlation inequality, was conjectured to be true by Tehranchi [19].

Conjecture 4. *Let $d > 0$ and let $K, T \subseteq \mathbb{R}^d$ be origin-symmetric convex sets, then*

$$\gamma_d(K + T)\gamma_d(K \cap T) \geq \gamma_d(K)\gamma_d(T) \quad (6)$$

where γ is the usual Gaussian probability measure.

In [19], Tehranchi showed that Conjecture 4 holds up to some constant, improving on a result by Schechtman, Schlumprecht, and Zinn [15] (these results and other related works are elaborated on in the Background section).

The proof of the equivalence of Conjectures 4 and 2 is a classical argument, which we present in Proposition 7 in the Background section. Using a similar argument we show that Theorem 3 implies Theorem 1.

We also present two partial results in the direction of the conjecture. First, we prove it for unconditional convex sets.

Theorem 5. *Let $K, T \subseteq \mathbb{R}^d$ be unconditional convex sets, then (6) holds.*

Figure 1: Source PDF page 2: Conjecture 2, Conjecture 4, and the source's unconditional-set partial theorem.

2 Literature input and exact implication

Recall that a positive function F on a product lattice is MTP_2 if

$$F(a \wedge b)F(a \vee b) \geq F(a)F(b) \quad (a, b \geq 0).$$

A symmetric positive definite matrix is an M -matrix here if its off-diagonal entries are nonpositive.

Theorem 1 (Karlin–Rinott consequence). *Let $X \sim N(0, \Sigma)$ be a nonsingular centered Gaussian vector. Suppose there is a diagonal sign matrix $S = \text{diag}(\pm 1, \dots, \pm 1)$ such that $S\Sigma^{-1}S$ is an M -matrix. Then for all $s, t \in [0, \infty]^n$,*

$$\mathbb{P}(|X| \leq s \wedge t) \mathbb{P}(|X| \leq s \vee t) \geq \mathbb{P}(|X| \leq s) \mathbb{P}(|X| \leq t). \quad (3)$$

Consequently, the strong Gaussian-box inequality (2) holds.

Proof. Karlin and Rinott [2] proved that the stated signed- M -matrix condition is precisely the condition under which the absolute-value vector $|X|$ is MTP_2 . Royen [3] restates this criterion explicitly on page 2 of the included supporting paper.

The lower-orthant distribution function

$$F(a) = \mathbb{P}(|X_i| \leq a_i \text{ for every } i)$$

is therefore MTP_2 . One may see the standard integration closure directly: the order kernel $\mathbf{1}_{\{x \leq a\}}$ is MTP_2 jointly in (x, a) ; products preserve MTP_2 ; and integration over x preserves MTP_2 . Thus

$$F(s \wedge t)F(s \vee t) \geq F(s)F(t),$$

which is (3). Finally $s + t \geq s \vee t$ coordinatewise and F is increasing, so $F(s + t) \geq F(s \vee t)$. This proves (2). Infinite coordinates follow by monotone convergence. $\square$

$$|\mathbf{I} + \Sigma \mathbf{T}|^{-\alpha} \quad (4)$$

with the identity matrix $\mathbf{I}$, the associated covariance matrix Σ (not to confuse with the covariances of the components), $\mathbf{T} = \text{Diag}(t_1, \dots, t_n)$, $t_i \geq 0$, and real "degrees of freedom" $\nu = 2\alpha \in \mathbb{N} \cup [(n-1)/2, \infty)$, see [5]. The $\Gamma_n(\alpha, \Sigma)$ -cdf belonging to (4) is here denoted by $G_\alpha(\mathbf{x}, \Sigma)$. The most comprehensive collection for integral and series representations and approximations for $\Gamma_n(\alpha, \Sigma)$ -cdfs is possibly found in the appendix of [2]. In [4] some examples are found of MTP₂-cdfs $G_\alpha(x, \Sigma)$. The $\Gamma_n(\frac{1}{2}, \Sigma)$ -distributed random vector $(X_1, \dots, X_n)$ is distributed as $(\frac{1}{2}Z_1^2, \dots, \frac{1}{2}Z_n^2)$, where $(Z_1, \dots, Z_n)$ has an $N_n(\mathbf{0}, \Sigma)$ -distribution. According to [3], the vector $(|Z_1|, \dots, |Z_n|)$, and consequently $(X_1, \dots, X_n)$, is MTP₂ iff there exists a sign matrix $\mathbf{S} = \text{Diag}(s_1, \dots, s_n)$, $s_i = \pm 1$, which generates an M-matrix $\mathbf{S}\Sigma^{-1}\mathbf{S}$, i.e. with only non-positive off-diagonal elements and with only non-negative off-diagonal elements in $\mathbf{S}\Sigma\mathbf{S}$. This is equivalent to the infinite divisibility of the $\Gamma_n(\frac{1}{2}, \Sigma)$ -distribution, see [1]. The Laplace transform in (4) shows the invariance of the corresponding $G_\alpha(\mathbf{x}, \Sigma)$ -cdf under the transformation $\Sigma \rightarrow \mathbf{S}\Sigma\mathbf{S}$. The MTP₂-property is generally not stable under convolutions. The most simple example of an MTP₂-cdf $G_{1/2}(\mathbf{x}, \Sigma)$ arises from a non-singular Σ with correlations $\rho_{ij} = a_i a_j$, $i \neq j$, all $a_i \in (-1, 1)$. (At most one $|a_i| = 1$ is admissible.) For this example the MTP₂-property holds for all α too.

2. The Inequality

The proof of the inequality in the following Theorem is based on the decrease of the ratio in (2). To make this paper more self-contained, the short proof for this decrease is given here again.

Proposition. If the cdf $F \in C^2(\mathbb{R}^n)$ (or $F \in C^2(\mathbb{R}_+^n)$) is strict MTP₂ with an everywhere positive density f on $\mathbb{R}^n$ (or f on $\mathbb{R}_+^n$), then the ratio

$$\frac{P(\bigcap_{i=1}^n A_i)}{P(\bigcap_{i \in I} A_i)P(\bigcap_{i \notin I} A_i)}, \quad \emptyset \neq I \subset \{1, \dots, n\},$$

is decreasing in every x_i .

Proof. Let e.g., i be an index of I . With the above assumption this decrease is equivalent to

$$\frac{\partial}{\partial x_i} \log P(\bigcap_{k=1}^n A_k) < \frac{\partial}{\partial x_i} \log P(\bigcap_{k \in I} A_k). \quad (5)$$

According to formula (1) it is

$$\frac{\partial}{\partial x_i} \left(\frac{\partial}{\partial x_i} \log P(\bigcap_{k=1}^n A_k) \right) > 0, \quad i \in I, j \notin I.$$

Since $\lim_{\min_{j \neq i} x_j \rightarrow \infty, j \notin I} P(\bigcap_{k=1}^n A_k) = P(\bigcap_{k \in I} A_k)$, the inequality (5) is satisfied. $\square$

The following probability inequality (6) is a consequence of the decreasing ratio in (2). It is proved inductively by successive convolutions. We write the vector $\mathbf{x}$ as $\mathbf{x} = \mathbf{x}_1 \oplus \mathbf{x}_2$, where $\mathbf{x}_1$ has the components x_k , $k \in I_1$, $\emptyset \neq I_1 \subset \{1, \dots, n\}$. The corresponding marginal cdfs of $F(\mathbf{x})$ are denoted by $F_i(\mathbf{x}_i)$ and the corresponding probability densities by $f_i(\mathbf{x}_i)$, $i = 1, 2$.

Figure 2: Royen, arXiv:2410.04143, page 2, restating the Karlin–Rinott signed- M -matrix characterization of MTP₂ absolute Gaussian vectors.

3 Geometric corollary

For an invertible matrix $A \in \mathbb{R}^{d \times d}$ and $a \geq 0$, put

$$P_A(a) = \{x \in \mathbb{R}^d : |(Ax)_i| \leq a_i \text{ for every } i\}.$$

Corollary 2. *Suppose $(AA^\top)^{-1}$ is sign-similar to an M -matrix. Then every pair $K = P_A(s)$ and $T = P_A(t)$ satisfies (1). In dimension two the matrix condition is automatic, so this includes every pair of centrally symmetric parallelograms having the same two facet directions.*

Proof. For a standard Gaussian Y , the vector AY has covariance $\Sigma = AA^\top$. Since A is bijective,

$$P_A(s) \cap P_A(t) = P_A(s \wedge t), \quad P_A(s) + P_A(t) = P_A(s + t).$$

Theorem 1 now gives (1). In dimension two a single coordinate sign flip can always make the sole off-diagonal entry of a positive definite precision matrix nonpositive. $\square$

4 Scope and full-upgrade audit

This does not settle the source conjecture. In three or more coordinates a precision matrix may contain a sign-frustrated cycle and fail to be sign-similar to an M -matrix. Indeed, numerical tests found that the stronger max/min inequality (3) can fail outside the MTP_2 class, even though the conjectured sum/min inequality continued to hold in the same tests.

Eight materially distinct full-upgrade routes were pursued: an orthogonal product-space section argument; MTP_2 and an elementary bivariate proof; the failed global max/min shortcut; deterministic two-dimensional polygon searches; Sobol searches over non-lattice three-dimensional polytopes; Ehrhard/Prékopa and radial reductions; and inverse Brascamp–Lieb plus tensorization. None closed the sign-frustrated general case or produced a counterexample. The detailed derivations, numerical scopes, and reusable scripts are recorded in the run attempt associated with this packet.

The supporting authors did not claim to answer Tehranchi’s conjecture. This packet records a direct implication of their MTP_2 theorem and is therefore classified as a literature-implied partial answer, with no novelty claim.

References

- [1] R. Assouline, A. Chor, and S. Sadovalsky, *A refinement of the Šidák–Khatrı inequality and a strong Gaussian correlation conjecture*, arXiv:2407.15684 (2024).
- [2] S. Karlin and Y. Rinott, *Total positivity properties of absolute value multinormal variables with applications to confidence interval estimates and related probability inequalities*, Ann. Statist. 9 (1981), 1035–1049.
- [3] T. Royen, *A probability inequality for convolutions of MTP_2 -distribution functions*, arXiv:2410.04143 (2024).